

IC Validator PERC Examples

5/8/2019

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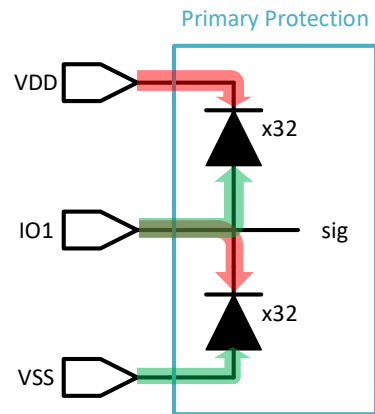
Current Path Checks

Point to Point Resistance and Current Density

Rule Notes

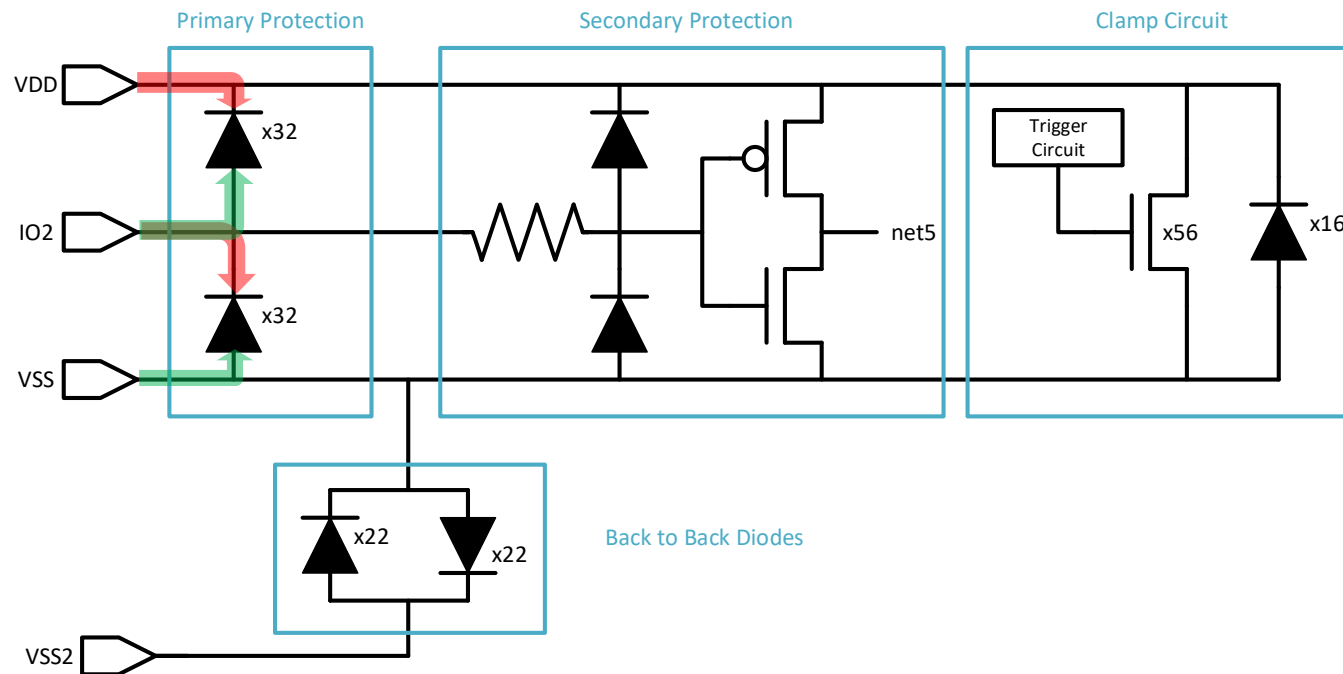
- Point to Point Resistance Rule Naming Convention
 - ESDRES[a][b][c] where:
 - a = [1 | 2] where 1 = primary current, 2 = secondary current
 - b = [1 | 2] where 1 = pad2pin path, 2 = pad2pad path
 - c = [r | b2b] where r indicates the secondary resistor is in the path, and b2b indicates back2back diodes
 - Example: “ESDRES21r”: 'secondary current', “I/O pad -> RESD -> CDM UP diode anode pin”
- Current Density Rule Names
 - ESDCD[1, 3, 4, 5] indicates a pad2pad path calculated with the primary current
 - ESDCD2 indicates a pad2pad path calculated with the secondary current
 - Example: “ESDCD3”: 'primary current', “Ground pad -> back-to-back diode -> Ground pad”
- Current Definitions
 - Primary current = HBM current
 - Secondary current = CDM current
- Rule definitions and constraints are found in the report_resistance_and_current_density() python function
- Constraint values are chosen to produce a fairly large number of errors without being overwhelming

ESDRES11 > 0.25 ohms

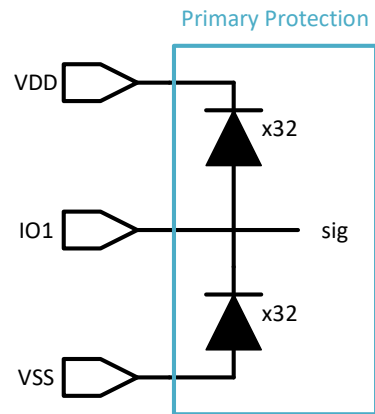


Check and report point-to-point resistance:

- From each I/O to the primary up protection diode(s) anode pin
- From each power pad to the primary up protection diode(s) cathode pin
- From each I/O to the primary down protection diode(s) cathode pin
- From each ground pad to the primary down protection diode(s) anode pin

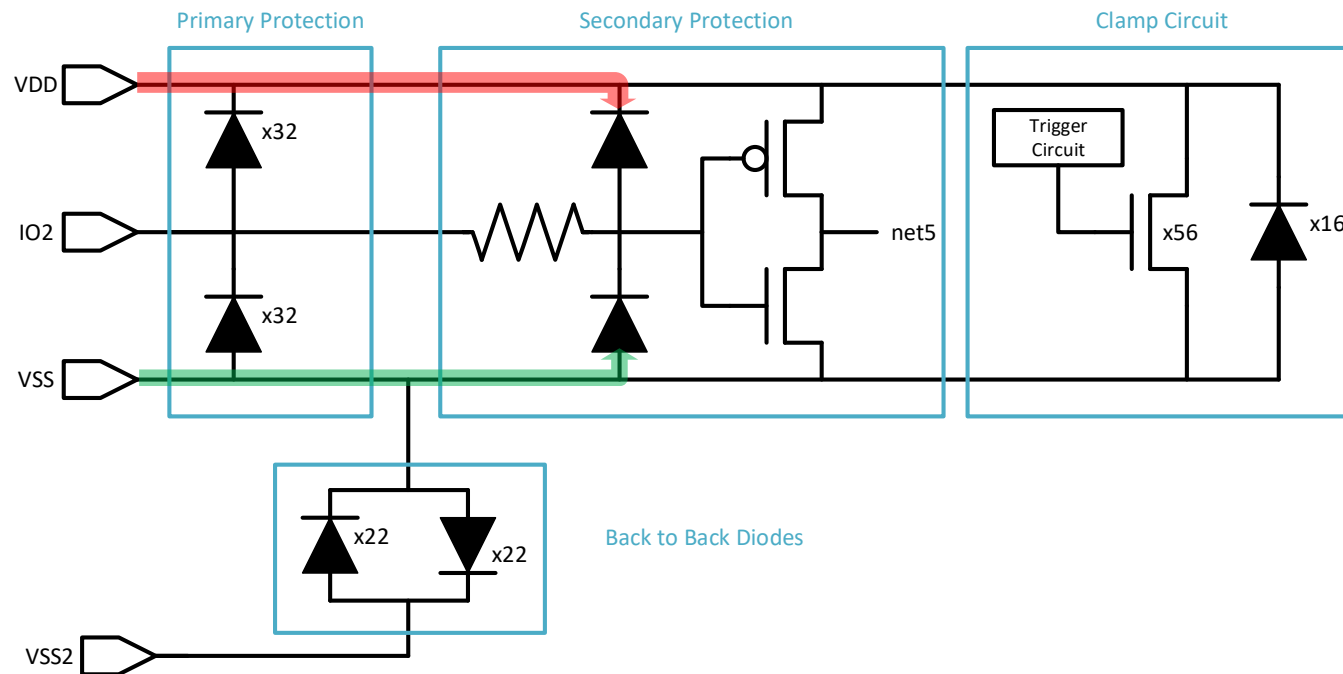


ESDRES21 > 2.5 ohms

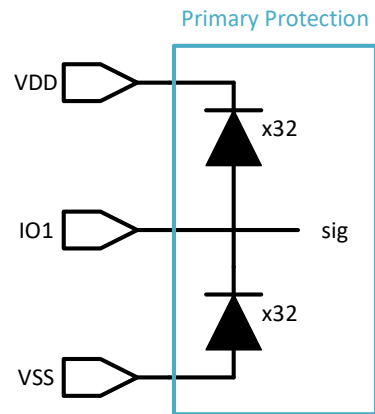


Check and report point-to-point resistance:

- From each power pad to the secondary up protection diode(s) cathode pin
- From each ground pad to the secondary down protection diode(s) anode pin

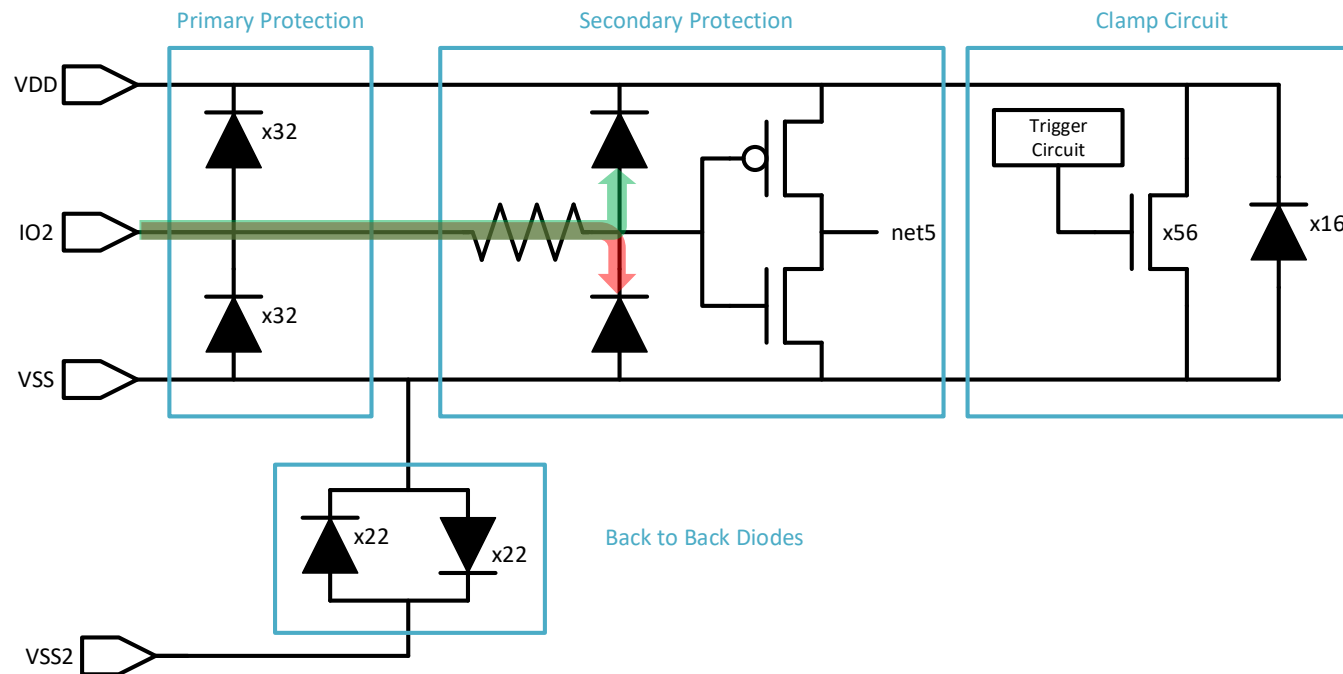


ESDRES21r > 666.5 ohms

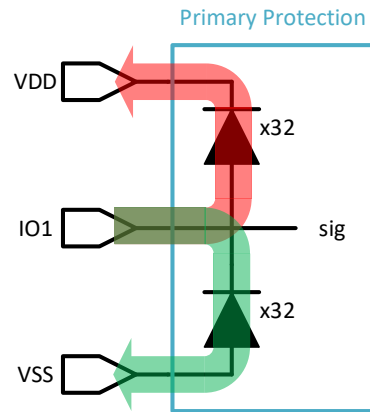


Check and report point-to-point resistance:

- From each I/O through the secondary resistor to the secondary up protection diode(s) anode pin
- From each I/O through the secondary resistor to the secondary down protection diode(s) cathode pin

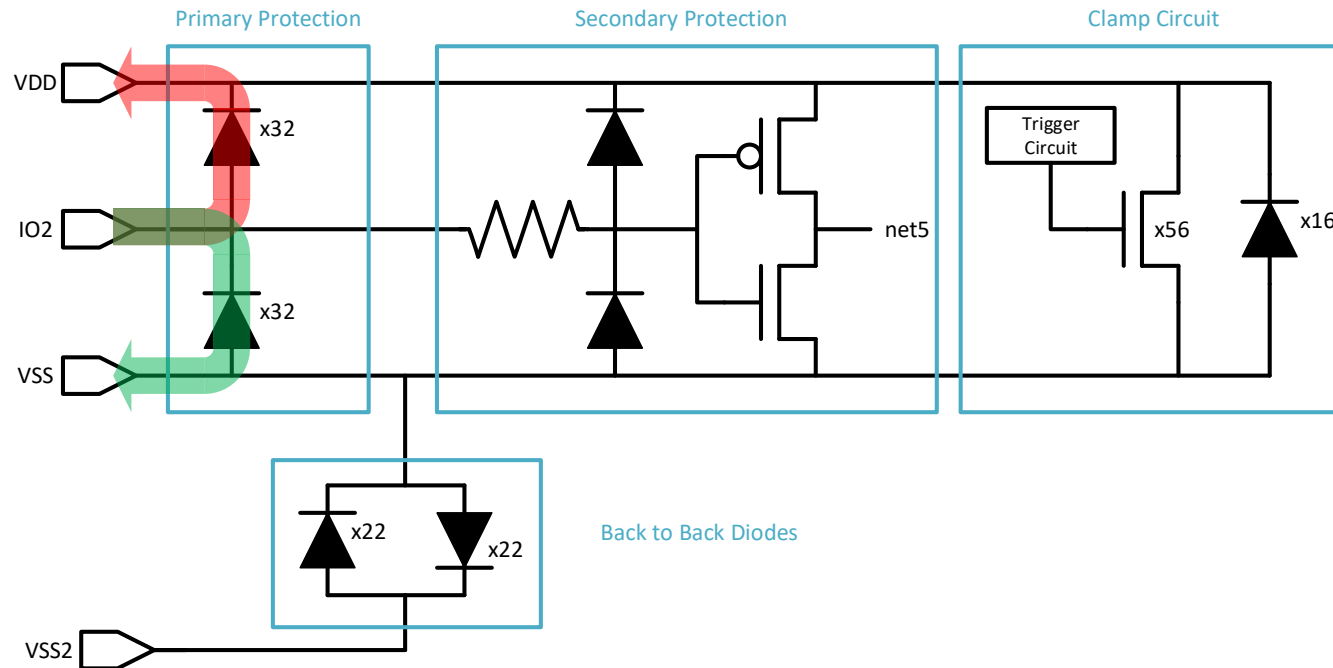


ESDCD1 / ESDRES12 > 0.4 ohms

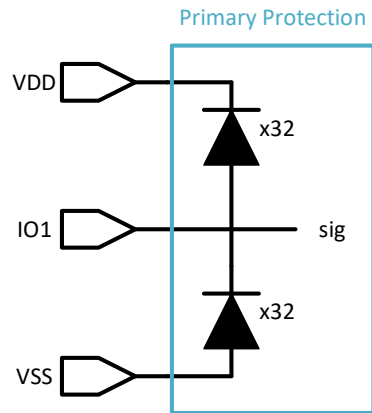


Check current density and report point-to-point resistance:

- From each I/O through the primary up protection diode(s) to the corresponding power pad
- From each I/O through the primary down protection diode(s) to the corresponding ground pad

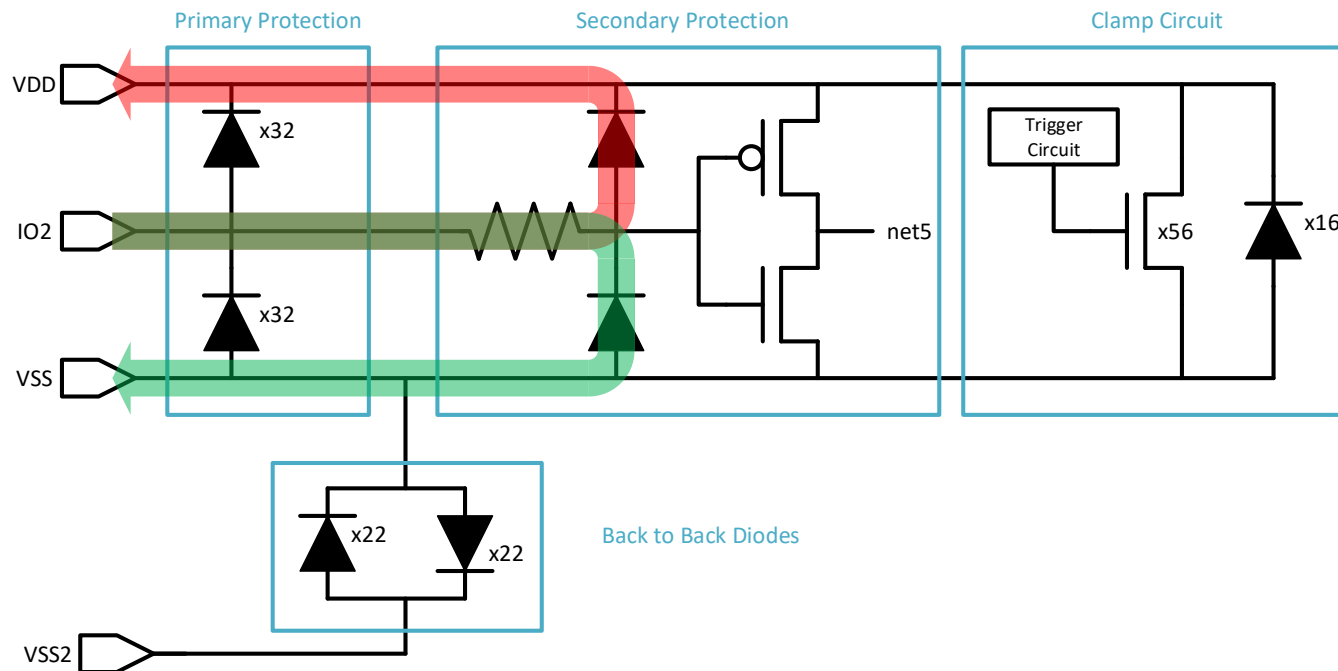


ESDCD2 / ESDRES22r > 666.5 ohms

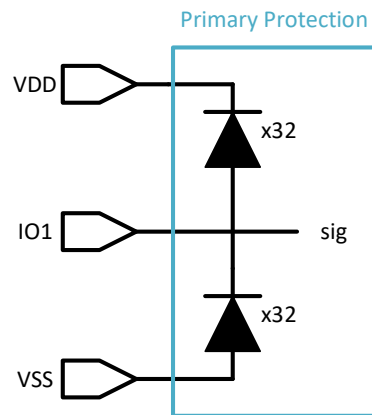


Check current density and report point-to-point resistance:

- From each I/O through the secondary series resistance and up protection diode(s) to the corresponding power pad
- From each I/O through the secondary series resistance and down protection diode(s) to the corresponding ground pad

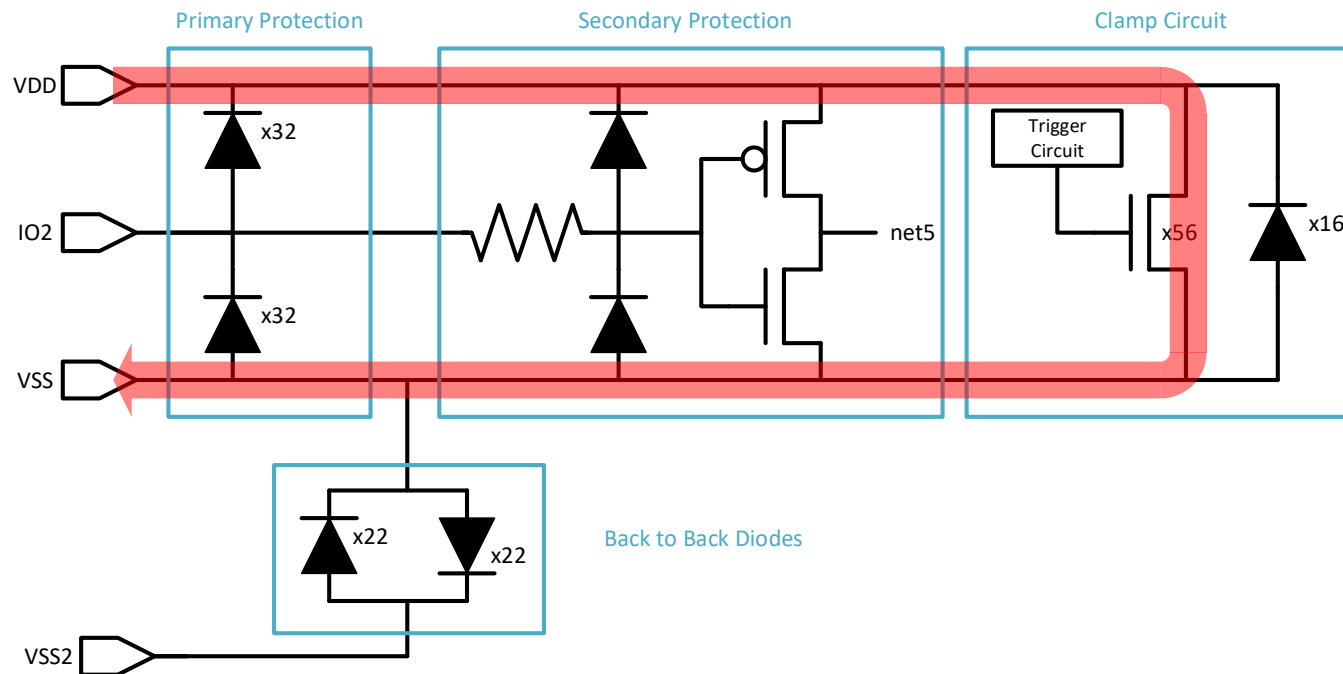


ESDCD5

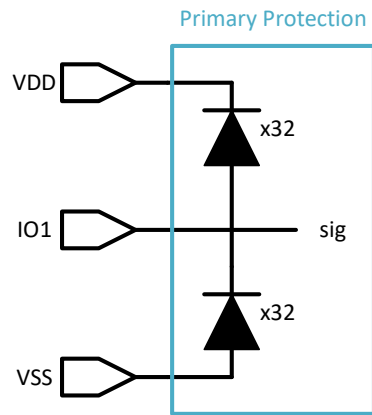


Check current density:

- For each nfet power clamp, from the corresponding power pad through the clamp source/drain to the corresponding ground pad

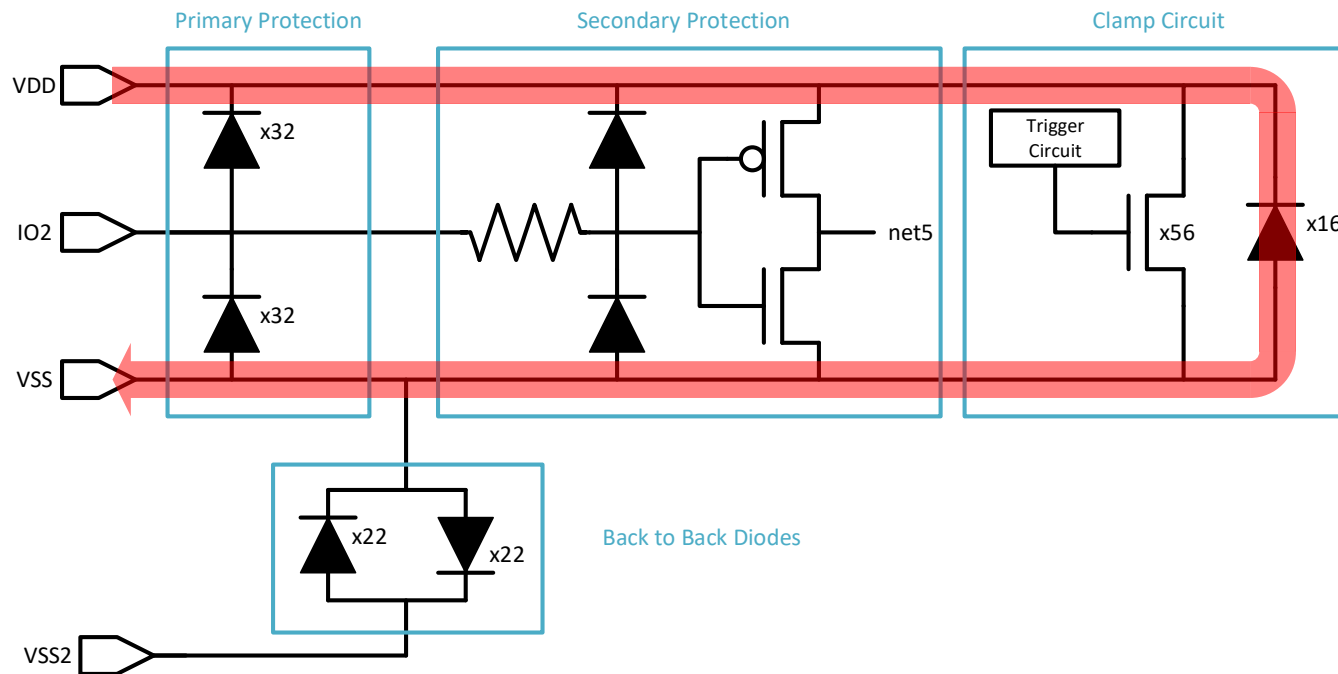


ESDCD3

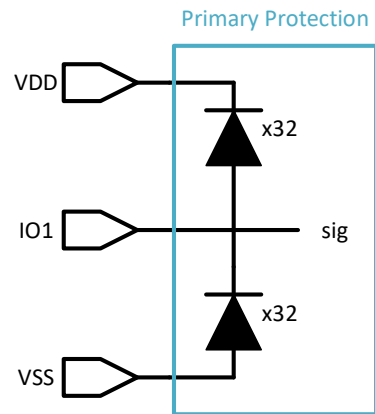


Check current density:

- For each power clamp reverse diode, from the corresponding power pad through the diode to the corresponding ground pad

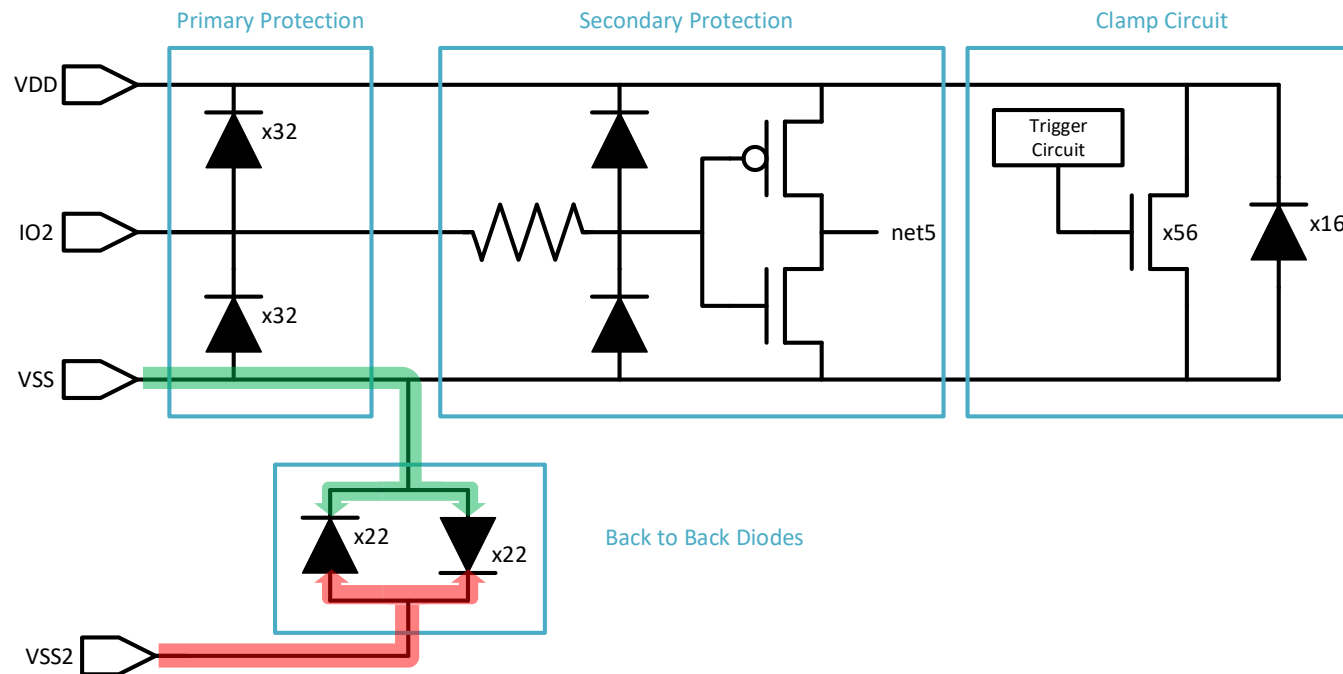


ESDRES11b2b > 0.1 ohms

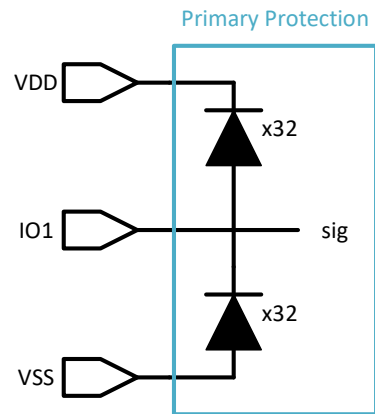


Report point-to-point resistance:

- For main ground (VSS)
 - From ground to back-to-back diode anode pin
 - From ground to back-to-back diode cathode pin
- For other ground (VSS2)
 - From other ground to back-to-back diode anode pin
 - From other ground to back-to-back diode cathode pin

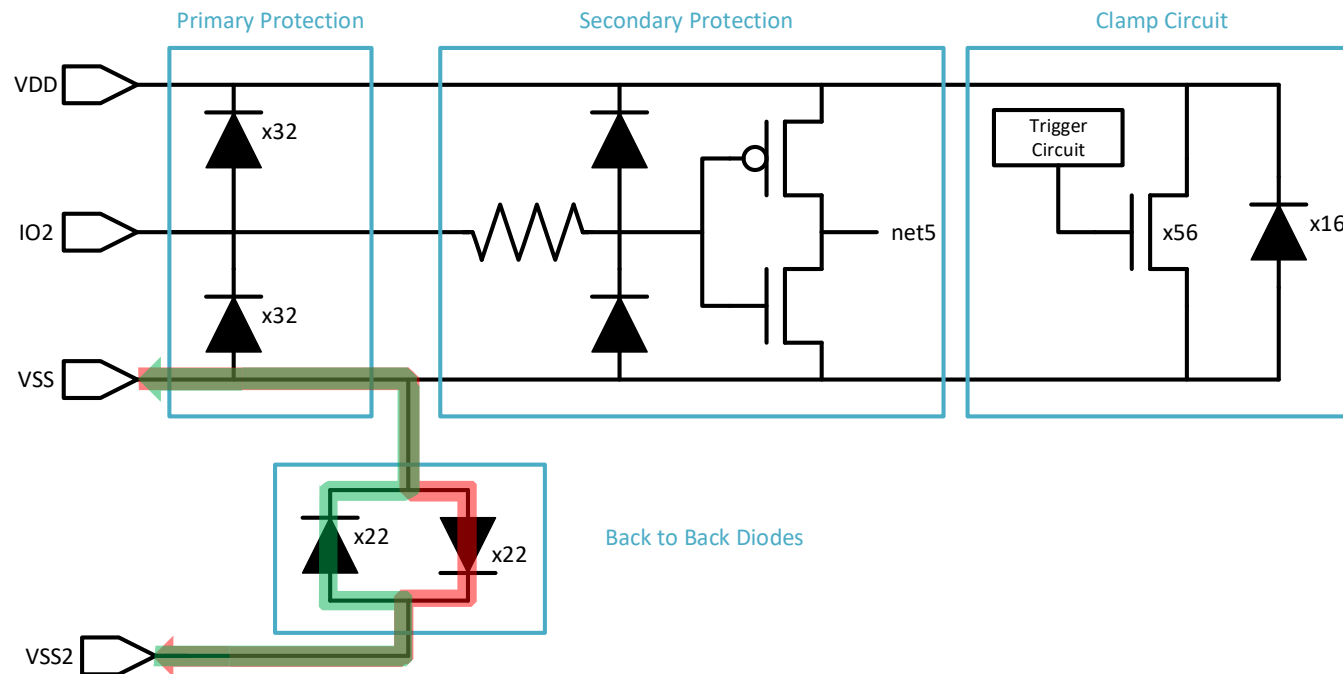


ESDCD4



Check current density:

- For each pair of main ground (VSS) to other ground (VSS2)
 - From main ground through one back-to-back diode to other ground
 - From other ground through the other back-to-back diode to main ground



ICV_VUE PERC Usage Hints

[illegible]

ICV_VUE PERC Usage Hints

The screenshot displays the Synopsys IC WorkBench Edit/View Plus interface for the ICV_VUE PERC project. The main window shows a layout view of the ICV_PERC_EXAMPLE. The left sidebar contains the 'Violation Browser' and 'Layout Layers' panels. The top menu bar includes File, View, Tools, Classification, Windows, and Help. The top toolbar contains various icons for navigation and analysis. The right sidebar contains the 'Highlight Path' panel.

Red callout boxes highlight the following features:

- Select Heat Map Data to Highlight:** Points to the 'Current Density' dropdown in the 'Highlight Path' panel.
- Select Specific Heat Map Bins:** Points to the color selection options in the 'Highlight Path' panel.
- Select Layers on the Path:** Points to the 'Layers (19)' list in the 'Highlight Path' panel.
- Select Nets on the Path:** Points to the 'Nets (2)' list in the 'Highlight Path' panel.
- Click to Highlight the Path in ICWBEV:** Points to the 'Highlight' button in the 'Highlight Path' panel.

The 'Violation Browser' panel shows the following error details:

Cell/Violation/Error Object/Placement	Error Count	Description
ICV_PERC_EXAMPLE	21	
ESDCD1: Primary Current Density	4	
IO1 -> VDD	1	I/O pad -> HBM UP diode a... cathode
IO2 -> VDD	1	I/O pad -> HBM UP diode a... cathode
IO1 -> VSS	1	I/O pad -> HBM DOWN diod... anode
	1	I/O pad -> HBM DOWN diod... anode

The 'Layout Layers' panel shows the following layers:

Layer	Type
nwell(1:0)	Conducting
diff(2:0)	Conducting
diff_block(2:1)	Conducting
diff_1p8v(2:2)	Conducting
diff_res(2:3)	Conducting
poly(3:0)	Conducting
poly_block(3:1)	Conducting
poly_res(3:2)	Conducting
pimp(4:0)	Conducting
nimp(5:0)	Conducting
sblock(6:0)	Conducting
cont(7:0)	Conducting
via1(7:1)	Conducting
via2(7:2)	Conducting
via3(7:3)	Conducting
via4(7:4)	Conducting
via5(7:5)	Conducting

The 'Diagnosis' panel shows the following error details:

ID	Obj	Type	Name	Notes
2	Err	Path		I=4000.0000 (mA),...y_diodes_pad2pad]
	Port	IO2		(Source) metal8 (242.7130, -89.1940)
	Pin	9BCD3542159/9BCD...21260_1/X1 ANODE		pdiff (258.8800, 147.5300)
	Pin	9BCD3542159/9BCD...260_1/X1 CATHODE		ndiff (245.0000, 147.5300)
	Pin	9BCD3542159/9BCD...21260_5/X1 ANODE		pdiff (258.8800, 123.9800)
	Pin	9BCD3542159/9BCD...260_5/X1 CATHODE		ndiff (245.0000, 123.9800)
	Pin	9BCD3542159/9BCD...1260_13/X1 ANODE		ndiff (245.0000, 161.9300)
	Pin	9BCD3542159/9BCD...60_13/X1 CATHODE		pdiff (258.8800, 165.5300)
	Pin	9BCD3542159/9BCD...21260_6/X1 ANODE		ndiff (245.0000, 165.5300)
	Pin	9BCD3542159/9BCD...260_6/X1 CATHODE		ndiff (245.0000, 165.5300)
	Pin	9BCD3542159/9BCD...1260_14/X1 ANODE		pdiff (231.1200, 127.5800)
	Pin	9BCD3542159/9BCD...60_14/X1 CATHODE		ndiff (217.2400, 127.5800)
	Pin	9BCD3542159/9BCD...21260_1/X1 ANODE		ndiff (245.0000, 109.5800)
	Pin	9BCD3542159/9BCD...260_1/X1 CATHODE		pdiff (258.8800, 151.1300)

The 'Diagnosis' panel also includes a table with the following columns: ID, Net, Layer, Error (%), CD, Limit, I (mA), Width (µm), Length.

ICV_VUE PERC Usage Hints

The screenshot displays the ICV_VUE PERC software interface. The main window shows the 'Violation Browser' on the left, listing various error types and their counts. The 'Diagnosis' table in the center provides detailed error information, including ID, Net, Layer, Error (%), CD, Limit, Width (μm), and Length. A red callout bubble points to the 'Diagnosis' table with the text: 'Right Click to Get This Menu'. Another red callout bubble points to the 'Highlight Path' dialog on the right with the text: 'Recalculate Heat Map Bin Ranges Is a Useful Function When Ranges Change'.

Violation Browser

Cell/Violation/Error Object/Placement	Error Count	Description
ICV_PERC_EXAMPLE	21	
ESDCD1: Primary Current Density	4	
IO1 -> VDD	1	I/O pad -> HBM UP diode a... cathode
IO2 -> VDD	1	I/O pad -> HBM UP diode a... cathode
IO1 -> VSS	1	I/O pad -> HBM DOWN diod... anode
IO2 -> VSS	1	I/O pad -> HBM DOWN diod... anode
ESDCD3: Clamp Diode Current Density	1	
ESDCD5: Clamp NMOS Current Density	1	
ESDCD4: B2B Diode Current Density	2	
ESDRES11b2b: B2B Primary Point to Point Resistance	2	
ESDRES22r: Secondary Ful...udes Secondary Resistor)	2	
ESDRES21r: Secondary Po...des Secondary Resistor)	1	
ESDRES21: Secondary Point to Point Resistance	1	
ESDCD2: Secondary Current Density	2	
ESDRES12: Primary Full Path Resistance	2	
ESDRES11: Primary Point to Point Resistance	3	

Diagnosis

ID	Obj	Type	Name	Notes
2	Err	Path		=4000.0000 (mA)...y_diodes_pad2pad]
		Port	IO2	(Source) metal8 (242.7130, -89.1940)
		Pin	9BCD3542159/9BCD...21260_1/X1 ANODE	pdiff (258.8800, 147.5300)
		Pin	9BCD3542159/9BCD...260_1/X1 CATHODE	ndiff (245.0000, 147.5300)
		Pin	9BCD3542159/9BCD...21260_5/X1 ANODE	pdiff (258.8800, 123.9800)
		Pin	9BCD3542159/9BCD...260_5/X1 CATHODE	ndiff (245.0000, 123.9800)
		Pin	9BCD3542159/9BCD...1260_13/X1 ANODE	pdiff (258.8800, 165.5300)
		Pin	9BCD3542159/9BCD...60_13/X1 CATHODE	ndiff (245.0000, 165.5300)
		Pin	9BCD3542159/9BCD...21260_6/X1 ANODE	pdiff (258.8800, 147.5300)
		Pin	9BCD3542159/9BCD...260_6/X1 CATHODE	ndiff (245.0000, 165.5300)
		Pin	9BCD3542159/9BCD...1260_14/X1 ANODE	pdiff (231.1200, 127.5800)
		Pin	9BCD3542159/9BCD...60_14/X1 CATHODE	ndiff (245.0000, 127.5800)
		Pin	9BCD3542159/9BCD...21260_1/X1 ANODE	pdiff (258.8800, 147.5300)
		Pin	9BCD3542159/9BCD...260_1/X1 CATHODE	ndiff (245.0000, 147.5300)
		Pin	9BCD3542159/9BCD...21260_2/X1 ANODE	pdiff (258.8800, 147.5300)
		Pin	9BCD3542159/9BCD...260_2/X1 CATHODE	ndiff (245.0000, 147.5300)

Highlight Path

Colored by...
Current Density Error (%)

Max: 751.8148
Min: -100.0000

☒ > 581.4519
☒ <= 581.4519
☒ < 411.0000
☒ < 411.0000

Check All
Uncheck All
Check Selected Only
Uncheck Selected Only
Recalculate Bin Ranges

Layers (9)
☒ metal1 Conducting
☒ metal2 Conducting
☒ metal3 Conducting
☒ metal4 Conducting
☒ metal5 Conducting
☒ metal6 Conducting

Nets (2/2)
☒ VDD
☒ IO2

Hide VUE Highlight

Thank You

